Validation and Publication Review

Portable Evidence Runtime · 11 November 2026 · Review of version 0.2

Assessment

The revised manuscript is suitable for public discussion as an explicitly bounded research prototype and preprint. Its evidence supports deterministic behavior on synthetic fixtures in one Linux environment. It does not yet support a peer-reviewed validation claim, general platform compatibility, verified human authorship, natural-language reasoning, or comparative superiority.

This is an engineering and editorial review performed during repository preparation. It is not independent peer review, a comprehensive security audit, or a legal opinion. Only one research manuscript was supplied; this companion document is a validation report.

Provenance and reproducibility

GitHub initially contained only a README at commit `c516411`. The implementation came from the author-supplied ZIP. Its SHA-256 matched the supplied receipt:

`c0b52e403684938467edad2cd147408bc994be9b3ab5db95bc15139b8537fa22`

Text extracted from the separately supplied PDF matched text extracted from the ZIP's manuscript PDF. The v0.2 manuscript revises that supplied source; it does not claim to be the unchanged attachment. The demonstration capsule and its root are unchanged.

Executed checks

Check	Local result	Scope
Python unittest discovery	29/29	25 original tests, three CLI tests and one cross-language boundary test
Shared boundary inputs within that test	14/14	Python/JavaScript decisions and supported-answer evidence agree
Strict JavaScript core	11/11	SHA-256, fixed fixture, refusal and embedded HTML checks
Authoring JavaScript core	10/10	Seals, supersession, pinned parent, ambiguity and JSON round trip
Synthetic experiment	15/15	Declared statuses, supports and expected source value/revision
Offline Chromium automation	13/13	Includes fresh-context transfer and UI update regressions
Both standalone HTML builds	Passed	Generated outputs match their source inputs
CLI build, verify, ask and refusal checks	Passed	Temporary outputs, retained anchor and process exit behavior
Manuscript/report export	Passed	Pandoc TXT and XeLaTeX PDF generation

The Chromium run used version 156.0.8078.79, Linux, and an offline browser context. Administrator policy blocked direct `file://` navigation; the probe therefore loaded the identical standalone HTML into memory. No remote page requests or JavaScript errors were observed. A

fresh browser context exercised export/import without shared browser storage. This is not a physical device transfer or proof of direct-file compatibility. Machine-readable observations are in `experiment/results/`.

Corrections made during review

- Aligned Python and JavaScript policy field-order enforcement, Unicode scalar validation, value-length accounting, safe revision bounds, and exact-key/ID end-of-string matching. Added shared boundary regression inputs.
- Cleared a previously supported strict-browser answer after an oversized import. Refreshed authoring answers after imports, revisions, and resetting the ledger.
- Made the declared relocation scenario actually copy and reread a capsule. Added expected answer-value and revision assertions to supported scenarios.
- Added a strict HTML build synchronization check. Strengthened release checks to compare both builds, paper copies, and recorded experiment outcomes.
- Added related work and explicit limits on historical novelty, authorship metadata, optional unanchored API calls, and the project-specific encoding.
- Made `core-papers/` the canonical Markdown source and rebuilt all requested formats; retained the original `paper/` paths as generated copies.

Publication limits and remaining work

1. Independently reproduce direct-file cold transfer on a second device and operating system, recording failures and user interventions.
2. Execute the proposed matched comparison before reporting effectiveness or superiority. Synthetic pass rates do not estimate real-world accuracy.
3. Extend conformance tests and security review before processing hostile or sensitive corpora. This review is not exhaustive fuzzing or formal proof.
4. Define authenticated anchor distribution and verifier distribution before claiming identity or malicious-publisher protection. A root shipped in the same editable HTML is a demo convenience, not an independent trust channel.
5. Select explicit software and manuscript licenses if reuse is intended. No license was supplied; public visibility alone does not grant a reuse license. A license was not chosen on the author's behalf.
6. Expand the related-work review and seek external review for venue submission.

The paper should be described publicly as a working bounded reference prototype with local synthetic results and a proposed external evaluation. That description matches the present evidence.